

SABIHA GOKCEN, Turkey

WMO: 170630

Lat: 40.898N

Lon: 29.303E

Elev: 99

StdP: 100.14

Time Zone: 3.00 (E03)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.2	-6.1	2.3	1.7	-4.8	2.5	2.5	12.3	2.7	11.1	5.0	3.8	30	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	32.2	21.1	31.1	21.1	30.0	21.0	23.8	28.1	23.2	27.6	22.6	26.9	5.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.5	25.7	21.9	16.8	25.5	21.1	15.9	25.1	71.7	28.3	69.4	28.0	66.9	26.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.3	8.4	-4.5	35.2	2.0	1.9	-5.9	36.6	-7.1	37.7	-8.2	38.7	-9.7	40.1
			-5.1	24.5	2.0	0.9	-6.6	25.2	-7.7	25.7	-8.8	26.2	-10.2	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	6.3	7.3	9.3	13.2	18.1	22.4	24.7	25.3	21.4	16.4	12.4	8.1	(6)
(7)		DBStd	7.41	4.11	4.12	3.79	3.59	3.20	2.60	1.94	1.92	2.62	3.12	3.69	3.83	(7)
(8)		HDD10.0	393	128	94	61	9	0	0	0	0	1	17	83	(8)	
(9)		HDD18.3	1742	372	309	281	159	43	2	0	6	74	181	317	(9)	
(10)		CDD10.0	2385	14	19	39	104	252	372	456	473	343	199	89	26	(10)
(11)		CDD18.3	692	0	0	0	4	37	124	197	214	98	14	3	1	(11)
(12)		CDH23.3	5103	0	0	2	34	233	865	1615	1749	549	50	6	1	(12)
(13)		CDH26.7	1566	0	0	0	2	47	248	539	602	124	3	0	0	(13)
(14)	Wind	WSAvg	3.9	3.7	4.0	3.9	3.8	3.7	4.0	4.6	4.7	4.1	3.8	3.3	3.5	(14)
(15)	Precipitation	PrecAvg	639	97	60	61	48	28	23	19	28	36	64	79	111	(15)
(16)		PrecMax	872	223	151	154	100	90	64	74	135	166	178	190	266	(16)
(17)		PrecMin	429	11	9	8	8	2	0	0	0	7	27	30	(17)	
(18)		PrecStd	130	49	30	39	26	18	17	18	32	37	40	37	51	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.3	22.0	26.8	30.1	33.2	34.2	34.0	32.0	27.1	24.1	20.9	(19)
(20)		MCWB	11.0	12.0	13.1	15.9	18.2	20.6	20.8	21.3	20.3	17.7	15.1	13.1	(20)	
(21)		2%	DB	16.0	17.0	19.8	24.0	27.9	31.0	32.1	32.2	29.7	24.8	22.0	17.2	(21)
(22)		MCWB	10.2	11.0	11.7	14.5	17.6	20.5	21.1	21.6	20.0	17.0	14.4	11.9	(22)	
(23)		5%	DB	14.2	15.1	17.8	21.8	26.0	29.2	30.9	31.0	28.0	23.0	20.0	15.9	(23)
(24)		MCWB	9.2	10.1	11.1	13.5	17.1	19.7	20.9	21.6	19.4	16.6	13.9	11.1	(24)	
(25)		10%	DB	12.8	13.8	15.8	19.9	24.0	27.9	29.7	30.0	26.2	21.2	18.0	14.0	(25)
(26)		MCWB	8.7	9.5	10.3	13.0	16.4	19.5	20.9	21.7	18.8	16.2	13.3	10.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	13.0	13.6	16.5	20.0	23.4	24.3	24.8	22.9	19.5	16.7	13.7	(27)
(28)		MCDWB	16.6	18.1	20.1	24.7	26.2	28.8	29.9	28.6	28.0	23.8	21.1	18.3	(28)	
(29)		2%	WB	10.5	11.5	12.4	15.2	18.7	22.3	23.5	24.2	21.8	18.5	15.4	12.6	(29)
(30)		MCDWB	14.9	15.8	18.0	22.0	24.7	26.8	28.2	28.2	26.0	22.3	20.1	16.8	(30)	
(31)		5%	WB	9.7	10.5	11.5	14.2	17.8	21.5	22.7	23.6	21.0	17.7	14.3	11.6	(31)
(32)		MCDWB	13.6	14.7	16.5	20.7	23.7	26.0	27.3	27.7	25.2	21.0	18.8	15.2	(32)	
(33)		10%	WB	8.8	9.7	10.6	13.1	17.1	20.8	22.2	23.0	20.2	16.9	13.5	10.5	(33)
(34)		MCDWB	12.2	13.6	15.0	18.5	22.9	25.6	26.7	27.1	24.6	20.1	17.4	13.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.1	6.3	7.8	9.3	9.0	9.1	9.5	9.4	8.8	7.4	7.1	6.3	(35)	
(36)		5% DB	MCDBR	8.1	8.6	11.0	12.4	11.6	11.2	11.1	10.5	11.0	9.8	9.2	7.9	(36)
(37)		MCWBR	4.5	4.4	5.3	5.6	4.6	4.0	3.8	3.3	3.9	4.5	4.2	4.5	(37)	
(38)		5% WB	MCDBR	7.6	8.2	9.6	11.1	10.0	9.5	9.4	8.8	9.1	7.8	8.1	7.2	(38)
(39)		MCWBR	4.4	4.4	5.0	5.3	4.4	3.9	3.8	3.0	3.6	4.0	4.2	4.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.392	0.428	0.468	0.434	0.423	0.427	0.434	0.423	0.404	0.374	0.365	(40)	
(41)		taud	2.343	2.247	2.158	2.064	2.238	2.312	2.314	2.293	2.308	2.333	2.361	2.346	(41)	
(42)		Ebn at Noon	773	803	815	807	847	855	848	830	813	782	754	735	(42)	
(43)		Edh at Noon	88	112	136	160	138	128	127	126	116	102	86	82	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.51	2.20	3.48	4.82	6.10	6.92	7.12	6.17	4.59	3.00	1.98	1.39	(44)	
(45)		RadStd	0.22	0.32	0.33	0.53	0.50	0.56	0.25	0.36	0.35	0.27	0.26	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	-98	N/A	+44

Nomenclature: See separate page